

Development of an Embedded HPEC Telemetry Unit and Windshield HUD for the TUKZIE Rev 0 Platform



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Abstract

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Chapter 1

Introduction

1.1 Background to the study

The global transition toward sustainable urban mobility has catalysed the development of lightweight, electric-drive vehicles capable of autonomous operation [1]. Within this context, the TUKZIE Flagship Programme seeks to engineer an energy-aware, secure vehicle platform for last-mile logistics and urban freight applications. The TUKZIE Rev 0, a 72V 2200W electric cargo trike, represents a testbed for integrating advanced telemetry and driver-assistance systems in resource-constrained environments.

Traditional vehicle telemetry systems have relied heavily on “Head-Down Displays” (HDDs) or post-processing of logged data, both of which present significant limitations. HDDs require drivers to divert their gaze from the road, contributing to cognitive overload and attention capture—a phenomenon where the driver’s focus shifts away from critical environmental cues [2, 3]. Post-processing delays maintenance interventions and prevents real-time decision-making.

Recent advances in High-Performance Embedded Computing (HPEC) have enabled edge-based signal processing in vehicular applications, allowing for real-time sensor fusion and on-vehicle analytics [4]. Simultaneously, Head-Up Display (HUD) technology has matured, offering a safer interface that projects critical information directly into the driver’s line of sight [5, 6]. However, the integration of these technologies—particularly for low-cost, three-wheeled electric vehicles in developing contexts—remains an open challenge.

This study addresses this gap by developing an integrated HPEC telemetry unit and windshield HUD for the TUKZIE Rev 0, enabling real-time ride-smoothness characterisation and driver feedback.

1.2 Objectives of this study

1.2.1 Problems to be investigated

Transportation agencies and fleet operators face significant challenges in monitoring road health and vehicle performance cost-effectively. Reference-grade profiling equipment requires specialist instrumentation and operation, which limits the frequency and coverage of surveys in resource-constrained settings [8, 7]. Standard in-car displays often contribute to cognitive overload and attention capture, where drivers divert their gaze from the road to process telemetry [13]. Furthermore, in resource-constrained environments, there is a gap for a modular, low-latency system capable of edge-based signal processing for ride characterisation while providing a safe, augmented-reality interface.

Specifically, this study investigates the following research questions:

1. How accurately and repeatably can a low-cost edge-processing system characterise ride smoothness in real time using inertial sensors on a 72V electric cargo trike?
2. Which HUD information hierarchy, layout, and update strategy can present safety-critical telemetry while limiting visual clutter and driver distraction?
3. What end-to-end latency, data-loss, and recovery performance can be achieved across sensor acquisition, edge processing, local logging, and 4G/LTE synchronisation?
4. Which time- and frequency-domain features most effectively separate road-induced vibration from repeatable motor and powertrain vibration?

1.2.2 Purpose of the study

The significance of investigating these problems lies in bridging the gap between expensive, proprietary road-profiling equipment and the need for cost-effective, real-time mon-

itoring solutions. By leveraging High-Performance Embedded Computing and low-cost inertial sensors, this study aims to provide a modular, low-latency system that delivers augmented-reality telemetry while ensuring driver safety.

The findings will directly inform the development of the TUKZIE Rev 0 platform and contribute to the broader field of sustainable urban mobility by demonstrating a replicable, open-architecture approach to vehicle telemetry. This work aligns with the UCT Radar Remote Sensing Group’s focus on advanced sensing and the TUKZIE Flagship Programme’s mandate for energy-aware, secure vehicle platforms.

Furthermore, this research addresses the practical needs of fleet operators in developing regions, where infrastructure monitoring budgets are constrained and traditional data collection methods are often infeasible. By enabling continuous, automated pavement condition assessment, the proposed system supports more efficient maintenance planning and improved road safety.

1.3 Scope and Limitations

1.3.1 Scope

This study focuses exclusively on the TUKZIE Rev 0, a 72V 2200W electric cargo trike manufactured by SA Motorcycles [16]. The system integrates the following components:

- A hardware module housing inertial measurement units (IMUs), air-quality sensors, and environmental sensors.
- A multithreaded software architecture for real-time sensor fusion, telemetry processing, and ride-smoothness indexing.
- An optical Head-Up Display subsystem projecting critical telemetry (speed, battery state of charge, and navigation) and safety alerts onto the trike’s windshield.
- A 4G/LTE communication link for central database synchronisation.

The project is aligned with the UCT Radar Remote Sensing Group’s focus on advanced sensing and specifically addresses the “ride-smoothness” mandate of TUKZIE by isolating road-induced vibration from nominal motor and powertrain vibration using statistical

feature extraction. The scope is limited to the TUKZIE Rev 0 platform and utilises low-cost inertial and environmental sensors processed at the edge using microcontroller and single-board computer platforms [7].

The software architecture follows a modular design, with separate threads for:

- High-frequency sensor polling (IMU at ≥ 100 Hz).
- Compute-heavy signal-processing algorithms (ride characterisation).
- Graphical HUD rendering.
- Networking stacks (MQTT/HTTP over 4G/LTE-M).

This separation ensures that no process blocks another [17].

1.3.2 Limitations

The following limitations are acknowledged:

- The study is limited to a single vehicle platform (TUKZIE Rev 0), which may affect generalisability to other vehicle types or configurations.
- Sensor calibration assumes nominal operating conditions; significant environmental variations (extreme temperatures and humidity) may affect accuracy.
- Edge-processing latency measurements are conducted in controlled test environments; real-world performance may vary with network conditions and electromagnetic interference.
- The HUD subsystem uses projected-display technology on the existing windshield, not full holographic or augmented-reality integration [5].
- Time constraints limit extensive field testing to urban routes only; rural and off-road conditions are not evaluated.
- The system is designed as a removable module to ensure that the vehicle can be quickly returned to a standard setup without obstructions [17].
- The 4G/LTE communication depends on network availability; data synchronisation may be delayed in areas with poor coverage.

1.4 Plan of development

This report is organised as follows:

Chapter 2: Literature Review—Reviews existing telemetry systems for electric vehicles, edge-computing architectures for vehicular applications, pavement condition-assessment methods, and Head-Up Display technologies. Key works in pavement roughness evaluation using inertial sensors [7, 18] and HUD design principles [5, 6, 19] are critically examined.

Chapter 3: System Design and Methodology—Presents the hardware architecture, sensor selection, and software framework for the integrated telemetry unit. The multithreaded architecture, sensor protocols (I2C, SPI, and UART), and edge-processing pipeline are detailed.

Chapter 4: Implementation—Details the fabrication process of the hardware module, HUD integration, and communication protocols. The PCB design, enclosure fabrication, and system assembly are described.

Chapter 5: Results and Discussion—Presents validation results for data ingestion accuracy, edge-processing latency analysis, ride-smoothness characterisation, and 4G/LTE data-synchronisation performance. Statistical analysis of sensor data and comparison with reference measurements are provided.

Chapter 6: Conclusions and Recommendations—Summarises the findings, evaluates the system against the stated objectives, and suggests future work, including potential integration of machine learning for predictive maintenance and extension to multi-vehicle fleet monitoring.

Chapter 2

Literature Review

2.1 Review scope and evaluation framework

The proposed system combines four fields that are often treated separately: pavement and ride-quality sensing, embedded signal processing, vehicular telemetry, and automotive Head-Up Display (HUD) design. The literature is therefore evaluated against the needs of the TUKZIE Rev 0 rather than only by reported classification accuracy. Relevant evidence must also address computational cost, sampling requirements, sensitivity to vehicle speed and mounting position, real-time latency, communication loss, and driver workload. This framework is important because a technique that performs well on a passenger car, smartphone, or laboratory simulator may not transfer directly to a lightweight three-wheeled cargo platform.

2.2 Pavement roughness and vehicle ride smoothness

Road roughness and ride smoothness are related but not interchangeable concepts. The International Roughness Index (IRI) is calculated from a measured longitudinal road profile using a standard quarter-car simulation [8]. It provides a transferable engineering measure when the profile measurement and calibration procedures are controlled. By contrast, an accelerometer mounted on a vehicle measures the combined response of the road input, tyres, suspension, chassis, payload, speed, sensor orientation, and mounting location. Consequently, raw vertical acceleration cannot be interpreted as IRI without a validated mapping and suitable reference measurements.

González et al. demonstrated that road profiles can be classified from simulated axle and body accelerations, establishing the value of vehicle-response measurements for roughness estimation [9]. The approach is attractive because it reuses vehicle dynamics rather than requiring a dedicated laser profiler. Its limitation is equally important: the inferred roughness depends on the assumed vehicle model and operating conditions. A model calibrated for a conventional four-wheeled vehicle cannot be transferred uncritically to the TUKZIE trike, whose wheel layout, suspension response, payload distribution, and low operating speed differ.

Smartphone and low-cost inertial sensing studies extend this response-based approach. Sattar et al. found that accelerometers and positioning sensors can support scalable road surface monitoring, but identified device placement, orientation, speed, sampling rate, and inconsistent ground truth as persistent sources of variation [10]. Alatoom and Smadi addressed cost and portability through an embedded framework using microcontroller and single-board computer platforms [7]. Their architecture is closer to the present project than a smartphone-only system because it permits controlled sensor mounting, deterministic sampling, and local processing. Nevertheless, published low-cost systems generally target passenger vehicles or generic road-anomaly detection; they do not establish a calibrated ride-smoothness measure for a three-wheeled electric cargo vehicle.

These findings support a deliberately bounded claim for this project. The TUKZIE unit should first produce a repeatable, vehicle-specific ride-smoothness index and event log. Correlation with IRI may be investigated when reference profile data are available, but the system should not label its output as IRI solely from uncalibrated chassis acceleration. Repeatability over repeated traversals, sensitivity to speed and payload, and agreement with a reference sensor or labelled route provide more defensible validation criteria.

2.3 Inertial signal processing and vibration separation

An IMU signal contains gravitational acceleration, orientation changes, road excitation, vehicle-body motion, motor and drivetrain vibration, sensor noise, and transient events. The processing pipeline must therefore preserve road-related information while removing components that would make comparisons between runs misleading. Orientation compensation or transformation into a vehicle-fixed frame is required before features from the nominally vertical axis are compared. Resampling and timestamp checks are also neces-

sary because feature values become unreliable when samples are irregular or missing.

The literature uses both time-domain and frequency-domain features. Root-mean-square acceleration, standard deviation, peak-to-peak amplitude, crest factor, jerk, and threshold counts are computationally inexpensive and interpretable. Frequency-domain energy and power spectral density can distinguish sustained narrow-band vibration from broadband road impacts. Bui et al. show that signal-processing-driven rough-road detection can be executed on an edge microcontroller, supporting the feasibility of local classification without continuous cloud processing [18]. However, threshold-based methods can overfit a particular vehicle and speed, while more complex classifiers can hide the physical reason for a detection and demand a larger labelled dataset.

For the TUKZIE platform, separation of road excitation from motor and powertrain vibration requires an empirical baseline. Stationary tests at several motor speeds can identify repeatable spectral components associated with the powertrain. Coast-down and repeated route tests can then show which components persist without powered traction and which vary with the road surface. Windowed statistics should be normalised or stratified by vehicle speed, since the same spatial road feature produces a different temporal excitation as speed changes. Rather than relying on one feature, a small interpretable feature vector combining vertical RMS acceleration, jerk, robust peak measures, and energy in selected frequency bands offers a defensible balance between sensitivity and computational cost.

Whole-body vibration standards provide useful terminology but do not replace road-profile measurement. ISO 2631-1 defines methods for evaluating human exposure to whole-body vibration [11]. Its frequency-weighted measures can inform a comfort-oriented index, whereas IRI remains a road-profile statistic. Reporting these as separate outputs would prevent the common error of treating passenger comfort, vehicle response, and pavement geometry as the same quantity.

2.4 Edge computing and real-time telemetry architecture

Edge computing moves computation closer to the sensors to reduce response time, bandwidth use, and dependence on a remote service [12]. These benefits directly match the TUKZIE use case: the HUD and safety alerts must continue to operate when cellular

coverage is poor, and transmitting every raw high-frequency sample would consume more bandwidth and energy than transmitting compact features and events. The embedded HPEC concept also enables several sensor streams to be timestamped and fused before data leave the vehicle [4].

The term “high performance” must nevertheless be interpreted relative to the platform. The relevant performance is not maximum throughput, but predictable completion of time-sensitive work within a constrained power and thermal budget. A multithreaded design can isolate high-rate sensor acquisition from feature extraction, HUD rendering, local storage, and networking, as required by the project brief [17]. Threads alone do not guarantee real-time behaviour: shared locks, blocking input/output, memory allocation, and unbounded queues can still cause missed samples or stale displays.

A producer–consumer architecture is therefore appropriate. The acquisition task should timestamp samples and place them into bounded buffers; processing tasks should consume complete windows; the HUD should read the most recent validated state without waiting for network activity; and the communication task should transmit queued summaries asynchronously. Overload behaviour must be explicit. Dropping an old display frame may be acceptable, whereas silently losing an IMU window or corrupting timestamps is not. Performance should be reported as distributions of acquisition jitter, processing time, display age, queue occupancy, and end-to-end latency rather than as an average alone.

2.5 HUD design, attention, and visual workload

HUDs reduce the need for large downward eye movements, but their position in the forward field of view does not make them automatically safe. Liu reported that HUD use changes attention demand and driving performance, showing that display location and cognitive processing must be evaluated together [2]. Smith et al. similarly found that conventional head-down-display assessment methods do not fully capture HUD behaviour and that both driving and secondary-task performance should be considered [14]. The design problem is therefore not simply to project more telemetry onto the windshield.

Windshield-display research identifies a design space involving location, visual depth, contrast, information density, and the relationship between graphics and the external scene [19]. Peripheral presentation can support awareness, but misaligned or excessive graphics may compete with hazards for attention [3]. Optical quality also depends on

the windshield and projection system; distortion, double images, ambient illumination, and viewing position affect legibility [5]. These constraints are especially relevant to a removable, low-cost projector mounted on a trike, where vibration and changing daylight are likely to be more severe than in an integrated production vehicle.

Complexity is a central risk. Currano et al. showed that HUD complexity influences driver perceptions of automated vehicles [15], while cognitive-load experiments on connected HUDs demonstrate that adding screens or information channels can increase mental demand [13]. The TUKZIE HUD should therefore use a stable information hierarchy: persistent speed and battery state, brief prioritised safety alerts, and navigation cues only when required. Decorative graphics, rapidly changing values, and simultaneous low-priority notifications should be excluded. The system should also adopt fail-safe behaviour when data are stale, for example by removing or clearly marking invalid values rather than continuing to display them as current.

Evaluation should combine technical and human-centred measures. Technical measures include rendering time, update rate, luminance or contrast under representative lighting, and the age of displayed telemetry. Human-centred evaluation can use task completion, missed-event counts, glance behaviour, perceived workload, and structured feedback during controlled low-risk trials. This is more defensible than asserting that a HUD is safer solely because it is head-up.

2.6 Cellular synchronisation and data integrity

Cellular communication extends local telemetry to fleet and maintenance applications, but coverage and delay are outside the embedded controller’s direct control. The architecture should consequently separate the real-time control path from remote synchronisation. HUD operation and local event detection must not wait for a 4G/LTE response. Each record should carry a monotonic sample timestamp, a route or session identifier, sensor-quality flags, and a sequence number so that the server can detect gaps and duplicates.

MQTT is suitable for compact publish–subscribe telemetry, whereas HTTP is useful for configuration, bulk upload, and conventional database interfaces. Protocol choice alone does not ensure reliability. A store-and-forward queue, bounded retry policy, acknowledgement strategy, and reconnection test are required. The appropriate performance measures are local data completeness, queued backlog, synchronisation delay, duplicate rate, and recovery after an induced outage. These measures directly address the project’s

remote data-synchronisation deliverable [17].

2.7 Synthesis and research gap

Existing studies establish that low-cost inertial sensors can identify road anomalies, edge devices can process sensor windows locally, and HUDs can reduce downward glances. They also expose three limitations. First, acceleration-based roughness estimates are strongly dependent on the vehicle, speed, mounting, and calibration. Second, edge architectures are often evaluated by algorithm accuracy or mean latency without reporting jitter, queue behaviour, and outage recovery. Third, HUD benefits diminish when visual complexity, optical limitations, and cognitive workload are ignored.

No reviewed work integrates these concerns for a removable telemetry and windshield-HUD module on a lightweight three-wheeled electric cargo vehicle operating under constrained power, computation, and cellular coverage. This project addresses that gap by combining controlled IMU mounting, vehicle-specific vibration baselining, interpretable edge features, non-blocking local processing, store-and-forward telemetry, and a minimal HUD information hierarchy. The contribution is therefore not merely the coexistence of a sensor and display. It is the design and validation of an integrated architecture whose ride-smoothness repeatability, latency, data integrity, and presentation safety can be measured explicitly.

Chapter 3

Methodology

This is what I did to test and confirm my hypothesis.

You may want to split this chapter into sub chapters depending on your design. I suggest you change the title to something more specific to your project.

This is where you describe your design process in detail, from component/device selection to actual design implementation, to how you tested your system. Remember detail is important in technical writing. Do not just write I used a computer give the computer specifications or the oscilloscopes part number. Describe the system in enough detail so that someone else can replicate your design as well as your testing methodology.

If you use or design code for your system, represent it as flow diagrams in text.

IMPORTANT: Include a motivation for your selection of an appropriate technique, or engineering or IT tool to solve your project problem. Discuss any limitations if appropriate.

Chapter 4

Results

These are the results I found from my investigation.

Present your results in a suitable format using tables and graphs where necessary. Remember to refer to them in text and caption them properly.

4.1 Simulation Results

4.2 Experimental Results

Chapter 5

Discussion

Here is what the results mean and how they tie to existing literature...

Discuss the relevance of your results and how they fit into the theoretical work you described in your literature review.

Chapter 6

Conclusions

These are the conclusions from the investigation and how the investigation changes things in this field or contributes to current knowledge...

Draw suitable and intelligent conclusions from your results and subsequent discussion.

Chapter 7

Recommendations

Make sensible recommendations for further work.

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Appendix A

Additional Files and Schematics

Add any information here that you would like to have in your project but is not necessary in the main text. Remember to refer to it in the main text. Separate your appendices based on what they are for example. Equation derivations in Appendix A and code in Appendix B etc.

IMPORTANT: Appendix A (see the table below) should provide a summary of how you have met the GAs associated with the course, and where in the report the evidence can be found.

If you have used AI in writing your report, you need to provide details in one of the appendices of how you used it.

If appropriate, in a subsequent appendix, provide a link to any GitHub repository where the code and additional materials for your project can be found.

A.1 Declaration of Artificial Intelligence Use

Artificial intelligence tools were used to assist with clarifying the organisation of this report, improving the alignment and consistency of its presentation, refining wording, and checking LaTeX formatting. All AI-assisted suggestions were reviewed by the author. The author remains responsible for the report's technical content, analysis, interpretations, and final editorial decisions.

A.1. DECLARATION OF ARTIFICIAL INTELLIGENCE USE

GA	Requirement	Justification and section in the report
1	Problem-solving	Description here
4	Investigations, experiments and data analysis	Description here
5	Use of engineering tools	Description here
6	Professional and technical communication (Long report)	Description here
8	Individual work	Description here
9	Independent learning ability	Description here

Appendix B

Addenda

B.1 Ethics Forms